

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Industry 4.0

Section Id :	640653134556
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653374473
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532040191 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED

EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536642788. ✓ Yes

6406536642789. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653374474

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532040192 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct category in Column B for the costs in Column A

Column A	Column B
(1) Cost of inspection of incoming products	(a) Shortage cost
(2) Obsolescence of product	(b) Ordering cost
(3) Loss of customer goodwill	(c) Carrying cost

Options :

6406536642790. ✓ (1): (b), (2): (c), (3): (a)

6406536642791. ✗ (1) : (c), (2) : (b), (3) : (a)

6406536642792. ✗ (1) : (a), (2) : (b), (3) : (c)

Question Number : 3 Question Id : 6406532040199 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun-fist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist bicuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406536642798. ✖ P1-S1, P2-S3, P3-S4

6406536642799. ✔ P1-S2, P2-S1, P3-S3

6406536642800. ✖ P1-S2, P2-S1, P3-S4

Question Number : 4 Question Id : 6406532040212 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following examples can be referred as cyclic pattern?

Options :

6406536642827. ✔ Economic recession

6406536642828. ✖ Increases Ceiling Fan sales during winters

6406536642829. ✖ Increased tourist footfall towards the end of the year

Sub-Section Number : 3

Sub-Section Id : 640653374475

Question Shuffling Allowed : No

Question Id : 6406532040208 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (5 to 6)

Question Label : Comprehension

Suppose, you are responsible for material purchase in an organization. You have demand forecast of a product for next four weeks which are 3, 4, 2 and 4 units respectively. You need to purchase one type of raw material to produce the final product to meet the above demand. Consider, one unit of the final product requires one unit of this raw material. Whenever an order is placed, your organization needs to incur a cost of \$4 irrespective of the quantity ordered. Purchasing cost of the raw material is \$2 per unit. Holding cost of an item not gone for production at the end of a week is \$0.5 per week. You can order a maximum of 5 units in a week and the warehouse has capacity to store upto 4 units of inventory in a week. Note that you need to meet the demand in every week.

Solve the above problem using dynamic programming (backward recursion) and answer the given sub-questions for this problem.

Sub questions

Question Number : 5 Question Id : 6406532040209 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What does the state variable represent?

Options :

6406536642817. ✖ How many units of the raw material needs to be purchased in a week

6406536642818. ✖ Ending inventory with the warehouse in each week

6406536642819. ✔ Beginning inventory with the warehouse in each week

Question Number : 6 Question Id : 6406532040210 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What does the action set represent?

Options :

6406536642820. ✔ How many units of the raw material needs to be purchased in a week

6406536642821. ✖ Ending inventory with the warehouse in each week

6406536642822. ✖ Beginning inventory with the warehouse in each week

Sub-Section Number :

4

Sub-Section Id :

640653374476

Question Shuffling Allowed :

Yes

Question Number : 7 Question Id : 6406532040205 Question Type : MSQ

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the cost(s) from the following list which contribute(s) to the shortage cost:

Options :

6406536642807. ✔ Loss of customer goodwill

6406536642808. ✖ Loss due to customer return

6406536642809. ✔ Cost of losing a customer

6406536642810. ✔ Loss of profit associated with not delivering the product

Question Number : 8 Question Id : 6406532040211 Question Type : MSQ

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the assumptions for single period newsvendor model.

Options :

6406536642823. ✔ Products are separable

6406536642824. ✖ Demand is deterministic

6406536642825. ✔ Inventory decision or deliveries are made before the demand is realized

6406536642826. ✔ Overstocking and Understocking costs are linear to the quantities

Sub-Section Number :

5

Sub-Section Id :

640653374477

Question Number : 9 Question Id : 6406532040206 Question Type : MCQ
Correct Marks : 1.5

Question Label : Multiple Choice Question

Match the following columns by selecting the correct name from Column B for the labels in column A which are depicted in the **'Figure: EOQ model'** below:

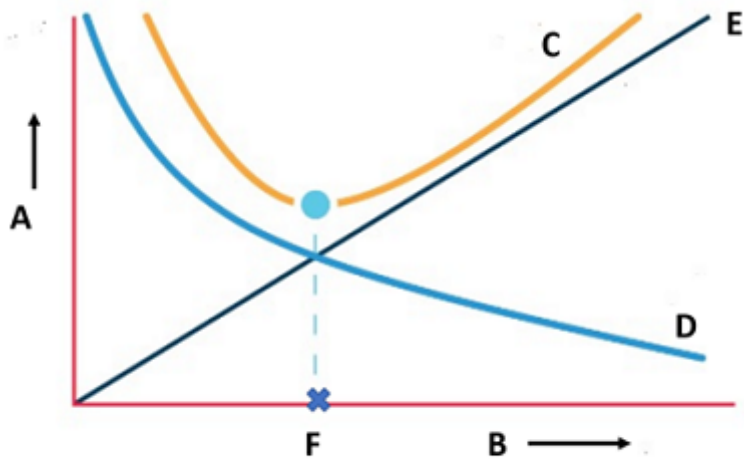


Figure: EOQ model

Column A	Column B
A	Carrying cost
B	Order cost
C	Cost
D	EOQ(Q*)
E	Quantity
F	Total cost

Options :

6406536642811. ✖ A:Cost; B:Quantity; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)
6406536642812. ✔ A:Cost; B:Quantity; C:Total cost; D:Order cost; E:Carrying cost; F:EOQ(Q*)
6406536642813. ✖ A:Quantity; B:Cost; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)
6406536642814. ✖ A:Total cost; B:EOQ(Q*); C:Cost; D:Order cost; E:Carrying cost; F:Quantity

Question Number : 10 Question Id : 6406532040207 Question Type : MCQ
Correct Marks : 1.5

Question Label : Multiple Choice Question

Which of the following model corresponds to Exponential Smoothing forecasting model?

Options :

$F_{t+1} = \alpha D_t + (1 - \alpha)F_t$ where F_t, D_t represents the forecast and actual data for period t

6406536642815. ✓

6406536642816. ✗ $F_t = \mu - \theta * \epsilon_{t-1}$ where $\epsilon_{t-1} = D_{t-1} - F_{t-1}$

Sub-Section Number : 6
Sub-Section Id : 640653374478
Question Shuffling Allowed : Yes

Question Number : 11 Question Id : 6406532040204 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

In a EOQ calculation, the estimated and actual parameter values are given in Table 1. What is the error in the EOQ calculation {error= $(Q_{\text{estimated}} - Q_{\text{actual}}) / Q_{\text{actual}}$ }. (Round your answer to two decimals without the percentage sign e.g, enter 10.12 if your answer is 10.12 %)

Table 1

Parameter	Estimate	Actual
Annual demand	1000 units	2000 units
Holding cost per unit per year	\$10	\$20
Ordering cost per order	\$50	\$25

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
40.00 to 42.00

Sub-Section Number : 7
Sub-Section Id : 640653374479
Question Shuffling Allowed : No

Question Id : 6406532040200 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (12 to 14)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 1 and earns Rs.2 by selling each Diya. Any Diya not sold by Diwali will be discounted to 50 Paise. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given subquestions:

Sub questions

Question Number : 12 Question Id : 6406532040201 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the Critical Ratio? (round off your answer to two decimal places)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.6 to 0.7

Question Number : 13 Question Id : 6406532040202 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

930 to 940

Question Number : 14 Question Id : 6406532040203 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the cost of overstocking is less than the cost of understocking, where will the optimal order quantity lie in the interval [800, 1000] units?

Options :

6406536642803. ✖ Toward left of the mean of uniform distribution [800,1000]

6406536642804. ✖ At the mean of uniform distribution [800,1000]

6406536642805. ✔ Towards right of the mean of uniform distribution [800,1000]

Sub-Section Number :

8

Sub-Section Id :

640653374480

Question Shuffling Allowed :

No

Question Id : 6406532040193 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (15 to 19)

Question Label : Comprehension

LIMAT Co. requires 1500 units of plastic ribbon per month. Each plastic ribbon costs Rs. 27 in the market. Placing an order costs Rs. 150 to Slipline Co. and the inventory carrying charges work out to 20% of the unit ribbon cost per year. Calculate the given subquestions:

Sub questions

Question Number : 15 Question Id : 6406532040194 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What should be the order quantity which minimizes the total cost to LIMAT Co.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1000

Question Number : 16 Question Id : 6406532040195 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the number of orders per year?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

18

Question Number : 17 Question Id : 6406532040196 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the annual order cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2700

Question Number : 18 **Question Id** : 6406532040197 **Question Type** : SA **Keyboard Layout** : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the average inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

500

Question Number : 19 **Question Id** : 6406532040198 **Question Type** : SA **Keyboard Layout** : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the annual inventory carrying cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2700